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Electronic device comprising display module

Abstract

An electronic device is provided. The electronic device includes a housing and a display including a transparent plate forming a front of the electronic device, a display panel viewable from the front of the electronic device through the transparent plate, a panel flexible circuit film arranged under the rear surface of the display panel including a bending part that extends toward the front surface of the display panel in a region adjacent to a first side surface of the electronic device, the bending part being electrically connected to the display panel on the front surface of the display panel, and a first support member arranged between the rear surface of the display panel and the panel flexible circuit film, a partial region of the first support member having an area larger than that of the panel flexible circuit film within a designated distance from the first side surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2021/012756, filed on Sep. 17, 2021, which is based on and claims the benefit of a Korean patent application number 10-2020-0122541, filed on Sep. 22, 2020, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2020-0181078, filed on Dec. 22, 2020, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(1) The disclosure relates to a display module structure including a member that supports a chip-on-film (COF) and an electronic device including the display module structure.

2. Description of Related Art

(2) Portable electronic devices (hereinafter, referred to as electronic devices) such as smart phones and tablet personal computers (PCs) may include a display module for displaying images. The display module may be manufactured by stacking a plurality of layers such as a display panel, a transparent plate, a cover panel, and the like in a manufacturing process and includes a display driver IC (DDI) and/or a chip-on-film (COF) configured to control the display module.

(3) In an electronic device, gaps (or empty spaces) may occur between respective components of the electronic device in process of manufacturing the electronic device. However, due to the gaps existing inside the display module or around the display module, when the electronic devices are dropped, the display module may receive a stronger impact, which may cause cracks in the display module.

(4) The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

(5) Recently, as the demand for high-performance display modules increases, a display module using a display driver IC (DDI) supporting an operating frequency of 120 hertz (Hz) or an electronic device to which the display module is applied is being released. In the case of the electronic device to which the 120 Hz DDI is applied, the size of a COF inevitably increases as the number of wires and/or packet assemblers/disassemblers (PADs) increases compared to that of a conventional electronic device to which a 60 Hz DDI is applied.

- (6) In the case of an electronic device to which an extended COF is applied, when the electronic device is dropped, cracks may more easily occur in the display module due to the gaps existing inside the electronic device (particularly, at the corners inside the electronic device).
- (7) Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device in which at least one support member configured to compensate for a gap and/or a step occurring inside the display module of the electronic device is disposed so that a display module can be suppressed from being damaged due to drop impact.
- (8) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
- (9) In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a housing defining at least a portion of an exterior of the electronic device, and a display seated on the housing. The display includes a transparent plate defining at least a portion of the front surface of the electronic device, a display panel at least partially visually recognized from the front surface of the electronic device through the transparent plate, a panel flexible circuit film including a bending portion disposed under the rear surface of the display panel and extending toward the front surface of the display panel in an area adjacent to a first side surface of the electronic device, wherein the bending portion is electrically connected to the display panel on the front surface of the display panel, and a first support member disposed between the rear surface of the display panel and the panel flexible circuit film. At least a partial area of the first support member may have an area larger than the area of the panel flexible circuit film within a predetermined distance from the first side surface.
- (10) In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes a display panel, a panel flexible circuit film including a bending portion disposed under the rear surface of the display panel and extending toward the front surface of the display panel in an area adjacent to a first side surface of the electronic device, wherein the bending portion is electrically connected to the display panel on the front surface of the display panel, and a first support member disposed between the rear surface of the display panel and the panel flexible circuit film. At least the partial area of the first support member may be exposed outside the panel flexible circuit film when the display is viewed from the rear surface of the display.
- (11) In the electronic device according to various embodiments of the disclosure, by extending the area of the support member disposed between the rear surface of the display panel and the panel flexible circuit film to the corners of the electronic device, it is possible to reduce empty spaces occurring inside the electronic device, and it is possible to suppress the occurrence of cracks in the display due to a drop impact.
- (12) In the electronic device according to various embodiments of the disclosure, by additionally disposing the support member configured to compensate for a step (or an empty space) occurring by a panel flexible circuit film on the display to reduce an empty space occurring inside the electronic device, it is possible to suppress cracks from occurring in the display due to a drop impact.
- (13) Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying

drawings, in which:

- (2) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;
- (3) FIG. 2A is a front perspective view of an electronic device according to an embodiment of the disclosure;
- (4) FIG. 2B is a rear perspective view of the electronic device of FIG. 2A according to an embodiment of the disclosure;
- (5) FIG. 3 is an exploded perspective view of the electronic device according to an embodiment of the disclosure;
- (6) FIGS. 4A, 4B, and 4C are cross-sectional views illustrating, in comparison, internal arrangement structures of an electronic device according to a comparative embodiment and electronic devices according to various embodiments of the disclosure;
- (7) FIGS. 5A and 5B are plan views illustrating, in comparison, internal arrangement structures of a display module according to a comparative embodiment and a display module according to various embodiments of the disclosure;
- (8) FIG. 6 is a view illustrating a state in which an on-cell-touch active matrix organic light emitting diode (AMOLED) (OCTA) module and a first support member are disposed on the rear surface of a display module according to an embodiment of the disclosure;
- (9) FIG. 7 is a view illustrating a state in which a second support member, a panel flexible circuit film, and an FPCB are disposed on the rear surface of the display module of FIG. 6 according to an embodiment of the disclosure;
- (10) FIG. 8A is a cross-sectional view of the display module of FIG. 7 taken in A-A' direction according to an embodiment of the disclosure;
- (11) FIG. 8B is a cross-sectional view of the display module of FIG. 7 taken in B-B' direction according to an embodiment of the disclosure;
- (12) FIG. 8C is a cross-sectional view of the display module of FIG. 7 taken in C-C' direction according to an embodiment of the disclosure;
- (13) FIG. 9 is a view illustrating impacts according to drop of an electronic device according to an embodiment of the disclosure;
- (14) FIG. 10 is a table showing, in comparison, impact distributions (or stress contours) due to the secondary impacts of falling of FIG. 9 in a drop simulation of electronic devices according to an embodiment of the disclosure;
- (15) FIG. 11A is a view illustrating assembling a first support member and the second support members to an OCTA module according to an embodiment of the disclosure;
- (16) FIG. 11B is a view illustrating assembling a panel flexible circuit film and an FPCB to the display module of FIG. 11A according to an embodiment of the disclosure; and
- (17) FIG. 11C is a view illustrating assembling a front housing to the display module of FIG. 11B according to an embodiment of the disclosure.
- (18) Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

DETAILED DESCRIPTION

(19) The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

(20) The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent

understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

(21) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

(22) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

(23) Referring to FIG. 1, an electronic device **101** in a network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(24) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(25) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g.,

by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(26) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(27) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(28) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(29) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(30) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(31) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(32) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(33) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(34) The connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**).

According to an embodiment, the connecting terminal **178** may include, for example, an HDMI

connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

(35) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(36) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(37) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(38) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(39) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a fifth generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(40) The wireless communication module **192** may support a 5G network, after a fourth generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the millimeter wave (mmWave) band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 gigabits per second (Gbps) or

more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(41) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(42) According to various embodiments, the antenna module **197** may form an mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(43) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(44) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102** and **104** or the server **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(45) The electronic device according to various embodiments may be one of various types of

electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(46) It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1.sup.st” and “2.sup.nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it denotes that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(47) As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(48) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. The term “non-transitory” simply denotes that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(49) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(50) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more

other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(51) FIG. 2A is a front perspective view of an electronic device according to an embodiment of the disclosure, and FIG. 2B is a rear perspective view of the electronic device of FIG. 2A according to an embodiment of the disclosure.

(52) Referring to FIGS. 2A and 2B, an electronic device **200** (e.g., the electronic device **101** in FIG. 1) according to an embodiment may include a housing **210** including a first surface (or a “front surface”) **210A**, a second surface (or a “rear surface”) **210B**, and a side surface (a “side wall”) **210C** surrounding the space between the first surface **210A** and the second surface **210B**. In another embodiment (not illustrated), the term “housing **210**” may refer to a structure defining a portion of the first surface **210A**, the second surface **210B**, and the side surface **210C** of FIGS. 2A and 2B.

(53) According to an embodiment, the first surface **210A** may be at least partially defined by a substantially transparent front surface plate **202** (e.g., a glass plate or a polymer plate including various coating layers). According to an embodiment, the front surface plate **202** may include a curved portion bent and seamlessly extending from the first surface **210A** toward a rear surface plate **211** in at least one side edge portion.

(54) According to an embodiment, the second surface **210B** may be defined by a substantially opaque rear surface plate **211**. The rear surface plate **211** may be made of, for example, coated or colored glass, ceramic, polymer, or metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of two or more of these materials. According to an embodiment, the rear surface plate **211** may include a curved portion bent and extending seamlessly from the second surface **210B** toward the front surface plate **202** in at least one side edge portion.

(55) According to an embodiment, the side surface **210C** may be defined by a side surface member (or a “bracket”) **218** coupled to the front surface plate **202** and the rear surface plate **211** and including metal and/or polymer. In some embodiments, the rear surface plate **211** and the side surface member **218** may be integrally configured and may include the same material (e.g., a metal material such as aluminum).

(56) According to an embodiment, the electronic device **200** may include at least one of a display **201**, an audio module **203**, a sensor module (not illustrated), camera modules **205**, **212**, **213**, and **206**, key input devices **217**, and a connector hole **208**. In some embodiments, in the electronic device **200**, at least one of the components (e.g., the key input devices **217**) may be omitted, or other components may be additionally included. In some embodiments, the electronic device **200** may include a sensor module (not illustrated). For example, the sensor module may be disposed on the rear surface of a screen display area of the display **201** that is visible to the outside of the electronic device **200** through the front surface plate **202**. For example, at least one of an optical sensor, an ultrasonic sensor, or a capacitive sensor may be disposed on the rear surface of the screen display area of the display **201**, but the disclosure is not limited thereto. In some embodiments, the electronic device **200** may further include a light-emitting element, and the light-emitting element may be disposed at a position adjacent to the display **201** in the area provided by the front surface plate **202**. The light-emitting element may provide, for example, the state information of the electronic device **200** in an optical form. In another embodiment, the light-emitting element may provide, for example, a light source that is interlocked with the operation of the camera module **205**. The light-emitting element may include, for example, an LED, an IR LED, and/or a xenon lamp.

(57) The display **201** may be visible to the outside of the electronic device **200** through, for example, a substantial portion of the front surface plate **202**. In an embodiment, the edges of the display **201** may be provided to be substantially the same as the outer peripheral shape (e.g., a curved surface) of the front surface plate **202** adjacent thereto.

(58) In another embodiment (not illustrated), the electronic device **200** may include a recess, a notch, or an opening in a portion of the screen display area of the display **201**, and may include other electronic components aligned with the recess, notch, or opening, such as the camera module **205** or a sensor module (not illustrated). In another embodiment (not illustrated), the rear surface of the screen display area of the display **201** may include at least one of the camera modules **212**, **213**, **214**, and **215**, a fingerprint sensor, and a flash (e.g., camera module **206**). In another embodiment (not illustrated), the display **201** may be coupled to or disposed adjacent to a touch-sensitive circuit, a pressure sensor capable of measuring a touch intensity (pressure), and/or a digitizer configured to detect an electromagnetic field-type stylus pen.

(59) The audio module **203** may include a microphone hole and a speaker hole. The microphone hole may include a microphone disposed therein to acquire external sound. In some embodiments, multiple microphones may be disposed in the microphone hole to detect the direction of sound. In some embodiments, the speaker hole and the microphone hole may be implemented as a single hole, or a speaker may be included without a speaker hole (e.g., a piezo speaker). The speaker hole may include an external speaker hole and a call receiver hole.

(60) By including a sensor module (not illustrated), the electronic device **200** may generate an electrical signal or data value corresponding to an internal operating state or an external environmental condition. The sensor module may further include, for example, a proximity sensor disposed on the first surface **210A** of the housing **210**, a fingerprint sensor disposed on the rear surface of the display **201**, and/or a biometric sensor (e.g., an HRM sensor) disposed on the second surface **210B** of the housing **210**. The electronic device **200** may further include at least one of sensor modules (not illustrated), such as a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(61) The camera modules **205**, **212**, **213**, **214**, **215**, and **206** may include a first camera device (e.g., camera module **205**) disposed on the first surface **210A** of the electronic device **200**, and second camera devices (e.g., camera modules **212**, **213**, **214**, and **215**) and/or a flash (e.g., camera module **206**) disposed on the second surface **210B** thereof. The above-described camera modules **205**, **212**, **213**, **214**, and **215** may include one or more lenses, an image sensor, and/or an image signal processor. The flash (e.g., camera module **206**) may include, for example, a light-emitting diode or a xenon lamp. In some embodiments, two or more lenses (e.g., an infrared camera lens, a wide-angle lens, and a telephoto lens) and image sensors may be disposed on one surface of the electronic device **200**.

(62) The key input devices **217** may be disposed on the side surface **210C** of the housing **210**. In another embodiment, the electronic device **200** may not include some or all of the above-mentioned key input devices **217**, and the key input devices **217**, which are not included in the electronic device **200**, may be implemented in another form, such as soft keys, on the display **201**. In some embodiments, the key input devices may include at least a portion of a fingerprint sensor disposed on the second surface **210B** of the housing **210**.

(63) The connector hole **208** may accommodate a connector configured to transmit and receive power and/or data to and from an external electronic device and a connector configured to transmit and receive an audio signal to and from an external electronic device. For example, the connector hole **208** may include a USB connector or an earphone jack. In an embodiment, the USB connector and the earphone jack may be implemented as a single hole (e.g., the connector hole **208** in FIGS. **2A** and **2B**), and according to another embodiment (not illustrated), the electronic device **200** may transmit and receive power and/or data or transmit and receive an audio signal to and from an

external electronic device (e.g., the electronic devices **102** and **104** in FIG. 1) without a separate connector hole.

(64) In the disclosure, the surface in the +z direction of FIGS. 2A and 2B will be referred to as the front surface, the surface in the -z direction of FIGS. 2A and 2B will be referred to as the rear surface, the end in the +y direction of FIGS. 2A and 2B will be referred to as the upper end, and the end in the -y direction of FIGS. 2A and 2B will be referred to as a lower end.

(65) FIG. 3 is an exploded perspective view of an electronic device according to an embodiment of the disclosure.

(66) Referring to FIG. 3, an electronic device **300** (e.g., the electronic device **200** of FIGS. 2A and 2B) according to an embodiment may include a display module **400** (e.g., the display **201** of FIG. 2A), a front housing **310** (e.g., the side surface member **218** of FIGS. 2A and 2B), a battery **320**, a main printed circuit board (PCB) **330**, and/or a rear surface plate **340** (e.g., the rear surface plate **211** of FIGS. 2A and 2B). The display module **400** according to an embodiment may include an on-cell-touch AMOLED (OCTA) module **410**, a first support member **420**, second support members **430**, a display driver IC (DDI) (or a display driving circuit) **440**, a panel flexible circuit film **450**, and/or a flexible printed circuit board (FPCB) **460**. At least one of the components of the electronic device **300** according to an embodiment may be the same as or similar to at least one of the components of the electronic device **101** of FIG. 1 and/or the electronic device **200** of FIGS. 2A and 2B, and redundant descriptions thereof will be omitted below.

(67) According to an embodiment, the on-cell-touch AMOLED (OCTA) module **410** may be a touch screen panel (TSP) in which touch sensor electrodes are directly deposited on an active organic light-emitting diode (OLED) display. The OCTA module **410** according to an embodiment may receive a user's touch input. According to an embodiment, the OCTA module **410** may include a transparent plate **411** (e.g., the front surface plate **202** of FIGS. 2A and 2B) configuring at least a portion of the front surface of the electronic device **300**, a display panel **412** including light-emitting pixels, and/or a cover panel **413** covering the rear surface of the display panel. According to an embodiment, four corners of the edges of the OCTA module **410** may have a round shape or a chamfered shape.

(68) According to an embodiment, the display panel of the OCTA module is configured to display an image and may include an OLED in which an organic light-emitting material may be disposed between at least two glasses and/or films. However, in various embodiments of the disclosure, the display panel is not limited to the OLED, and various types of display panels may be used.

(69) According to an embodiment, the first support member **420** may be disposed between the rear surface of the OCTA module **410** and the panel flexible circuit film **450** adjacent to the lower end of the electronic device **300**. The first support member **420** according to an embodiment may fill an empty space which occurs between the OCT module **410** and the panel flexible circuit film **450** by the display driver IC **440** and/or a bending portion (not illustrated) of the panel flexible circuit film **450**. According to an embodiment, the first support member **420** may be made of a durable material to protect the display driver IC **440**.

(70) According to an embodiment, the second support members **430** may each be disposed to cover a portion of the rear surface of the first support member **420**. The second support members **430** according to an embodiment may each fill an empty space, which may occur between the panel flexible circuit film **450** and the front housing **310** by the panel flexible circuit film **450**. According to an embodiment, the second support members **430** may be made of tape so that it can be bonded to the rear surface of the first support member **420**. According to an embodiment, a pair of second support members **430** may be respectively disposed at opposite ends of the first support member **420**.

(71) According to an embodiment, the panel flexible circuit film **450** may be disposed adjacent to the lower end portion of the electronic device **300** to cover at least a portion of the rear surface of the OCTA module **410** and may be disposed between the first support member **420** and the front

housing **310**. According to an embodiment, the panel flexible circuit film **450** may electrically connect the display panel **412** and the FPCB **460** to each other and may include the display driver IC **440** (or a display drive circuit) capable of controlling the display panel **412**.

(72) According to an embodiment, the display driver IC **440** may be electrically connected to the panel flexible circuit film **450** and may be disposed between the rear surface of the OCTA module **410** and the panel flexible circuit film **450**. According to an embodiment, the display driver IC **440** may receive image information including an image control signal corresponding to a command for controlling image data from the FPCB **460** and/or the main PCB **330** and may control the display panel **412** in response to the received image information to implement an image. For example, the display driver IC **440** may support an operating frequency of 120 Hz.

(73) According to an embodiment, the display driver IC **440** and the panel flexible circuit film **450** may configure a chip-on-film (COF) that controls the display module **400** (or the display panel **412**).

(74) According to an embodiment, the FPCB **460** may be disposed adjacent to the lower end portion of the electronic device **300** to cover at least a portion of the rear surface of the OCTA module **410** and/or at least a portion of the rear surface of the panel flexible circuit film **450**. According to an embodiment, the FPCB **460** may be electrically connected to the main PCB **330** of the electronic device **300**, and a plurality of electronic components (e.g., a processor, a memory, and an interface) may be disposed on the FPCB **460**.

(75) According to an embodiment, the front housing **310** may define at least a portion of the exterior of the electronic device **300** and may include a seating portion to be used to place various electronic components (e.g., the battery **320** and the main PCB **330**) thereon.

(76) According to an embodiment, the battery **320** may be disposed inside the front housing **310**. The battery **320** is a device for supplying power to at least one electronic component of the electronic device **300**, and may include, for example, a non-rechargeable primary battery, a rechargeable secondary battery, or a fuel cell. At least a portion of the battery **320** may be disposed substantially on the same plane as, for example, the main PCB **330**. According to an embodiment, the battery **320** may be integrally disposed within the electronic device **300**, or may be removably mounted on the electronic device **300**.

(77) According to an embodiment, the main PCB **330** may be accommodated and disposed inside the front housing **310**. According to an embodiment, a plurality of electronic components (e.g., a processor, a memory, and an interface) may be disposed on the main PCB **330**. The processor may include at least one of, for example, a central processor, an application processor, a graphics processor, an image signal processor, a sensor hub processor, or a communication processor. The memory may include, for example, volatile memory or nonvolatile memory. The interface may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may electrically or physically connect, for example, the electronic device **200** to an external electronic device and may include a USB connector, an SD card/an MMC connector, or an audio connector.

(78) According to an embodiment, the rear surface plate **340** may define the rear surface (e.g., the second surface **210B** of FIG. 2B) of the electronic device **300**. The rear surface plate **340** may protect the internal components of the electronic device **300** from an external impact or inflow of foreign substances.

(79) FIGS. 4A, 4B, and 4C are cross-sectional views illustrating, in comparison, internal arrangement structures of an electronic device according to a comparative embodiment and electronic devices according to various embodiments of the disclosure.

(80) FIG. 4A illustrates a display module **400a** and a front housing **310a** in an electronic device **300a** according to a comparative embodiment, FIG. 4B illustrates a display module **400b** and a front housing **310b** in an electronic device **300b** according to an embodiment of the disclosure, and FIG. 4C illustrates a display module **400c** and a front housing **310c** in an electronic device **300c**

according to an embodiment of the disclosure. For example, the display module **400a** of FIG. 4A and the display module **400b** of FIG. 4B may use a display driver IC that supports an operating frequency of 60 Hz, and the display module **400c** of FIG. 4C may use a display driver IC that supports an operating frequency of 120 Hz.

(81) Referring to FIG. 4A, in the electronic device **300a** according to the comparative embodiment, a first support member **420a** disposed between a panel flexible circuit film **450a** and the cover panel **413** of the OCTA module **410** may be disposed not to be exposed outside the panel flexible circuit film **450a**. That is, the first support member **420a** may be disposed not to extend to the end of the cover panel **413**. Accordingly, an empty space A inevitably occurs between the panel flexible circuit film **450a** and the cover panel **413**.

(82) In addition, in the electronic device **300a** according to the comparative embodiment, the front housing **310a** may include three steps **311a**, **312a**, and **313a** configured to accommodate the display module **400a**. For example, the front housing **310a** of FIG. 4A may include a first step **311a** on which the transparent plate **411** is seated, a second step **312a** on which the display panel **412** and the cover panel **413** are seated, and a third step **313a** on which the first support member **420a** and the panel flexible circuit film **450a** are seated. For example, the front housing **310a** of FIG. 4A may include the first step **311a** providing a space configured to accommodate the transparent plate **411**, the second step **312a** providing a space configured to accommodate the display panel **412** and the cover panel **413**, and the third step **313a** providing a space configured to accommodate the first support member **420a** and the panel flexible circuit film **450a**.

(83) Referring to FIG. 4B, in the electronic device **300b** according to an embodiment of the disclosure, a first support member **420b** disposed between a panel flexible circuit film **450b** and the cover panel **413** of the OCTA module **410** may be disposed to extend to an end of the panel flexible circuit film **450b**. For example, the first support member **420b** may be disposed to extend to the end of the panel flexible circuit film **450b**, but not be exposed outside the panel flexible circuit film **450b**. Accordingly, an empty space may not occur between the panel flexible circuit film **450b** and the cover panel **413**.

(84) According to an embodiment, the front housing **310b** in the electronic device **300b** of FIG. 4B may be the same as the front housing **310a** (e.g., the front housing **310a** including three steps **311a**, **312a**, and **313a** configured to accommodate the display module **400a**) in the electronic device **300a** of FIG. 4A.

(85) Referring to FIG. 4C, in the electronic device **300c** according to an embodiment of the disclosure, a first support member **420c** disposed between a panel flexible circuit film **450c** and the cover panel **413** of the OCTA module **410** may be disposed to be exposed outside the panel flexible circuit film **450c**. That is, the first support member **420c** may be disposed to extend (or expand) to the end of the cover panel **413**. Accordingly, when compared to FIG. 4A, it may be possible to eliminate the empty space between the panel flexible circuit film **450c** and the cover panel **413**.

(86) The display module **400c** of the electronic device **300c** according to an embodiment of the disclosure may further include second support members **430** disposed between the first support member **420c** and the front housing **310c**. According to an embodiment, the second support members **430** may each cover at least a portion of the rear surface of the first support member **420c** that is exposed outside the panel flexible circuit film **450c**. According to an embodiment, the second support members **430** may each fill an empty space that may occur between the first support member **420c** and the front housing **310c**, thereby minimizing empty spaces inside the electronic device **300c**.

(87) In addition, referring to FIG. 4C, in the electronic device **300c** according to an embodiment of the disclosure, the front housing **310c** may include two steps **311c** and **312c** configured to accommodate the display module **400c**. For example, the front housing **310c** of FIG. 4C may include a first step **311c** on which the transparent plate **411** is seated, and a second step on which the display panel **412**, the cover panel **413**, the first support member **420c**, the second support

members **430**, and the panel flexible circuit film **450c** are seated. For example, the front housing **310c** of FIG. **4C** may include the first step **311c** providing a space configured to accommodate the transparent plate **411**, and a second step providing a space configured to accommodate the display panel **412**, the cover panel **413**, the first support member **420c**, the second support members **430**, and the panel flexible circuit film **450c**. According to an embodiment, in the case of the electronic device **300c** according to an embodiment of the disclosure, the number of steps in which the display module **400c** is accommodated in the front housing **310c** may be reduced compared to the electronic device **300a** of FIG. **4**.

(88) In the electronic device **300c** according to an embodiment of the disclosure, by minimizing empty spaces around the display module **400c** inside the electronic device **300c** and the number of steps in which the display module **400c** is accommodated in the front housing **310c**, it is possible to suppress the display module **400c** from being damaged by a drop impact of the electronic device **300c**.

(89) FIGS. **5A** and **5B** are plan views illustrating, in comparison, the inner arrangement structures of a display module according to a comparative embodiment (e.g., a display module of FIG. **4A**) and a display module (e.g., a display module of FIG. **4C**). according to various embodiments of the disclosure

(90) FIG. **5A** illustrates a panel flexible circuit film **450'** and the FPCB **460** in the display module **400'** according to the comparative embodiment, and FIG. **5B** illustrates a panel flexible circuit film **450** and the FPCB **460** in the display module **400** according to an embodiment of the disclosure. For example, the display module **400'** of FIG. **5A** may use a display driver IC (not illustrated) that supports an operating frequency of 60 Hz, and the display module **400** of FIG. **5B** may use the display driver IC **440** that supports an operating frequency of 120 Hz.

(91) According to an embodiment, the panel flexible circuit films **450'** and **450** in FIGS. **5A** and **5B** may electrically connect the FPCB **460** and the OCTA module **410** to each other.

(92) When comparing FIGS. **5A** and **5B**, the panel flexible circuit film **450** using the 120 Hz display driver IC **440** according to an embodiment of the disclosure may cover the area of the rear surface of the OCTA module **410** greater than the area of the panel flexible circuit film **450'** using a 60 Hz display driver IC according to the comparative embodiment.

(93) Referring to FIG. **5A**, the panel flexible circuit film **450'** of the display module **400'** may not extend to opposite corners of the lower end portion of the OCTA module **410**, but may be disposed to cover only the central area of the lower portion of the OCTA module **410**. In contrast, in FIG. **5B**, the panel flexible circuit film **450** of the display module **400** of the disclosure may be disposed to cover the OCTA module **410** from the center area of the lower end portion of the OCTA module **410** to areas adjacent to opposite corners of the OCTA module **410**. According to an embodiment, in the case of the display module **400** using a 120 Hz display driver IC **440**, a panel flexible circuit film **450** having an area greater than that of the display module **400'** using a 60 Hz display driver IC may be used.

(94) Referring to FIGS. **6**, **7**, and **8A** to **8C**, an internal arrangement structure of a display module of an electronic device according to an embodiment of the disclosure will be described in detail.

(95) FIG. **6** is a view illustrating a state in which an OCTA module and a first support member are disposed on the rear surface of a display module according to an embodiment of the disclosure.

(96) Referring to FIG. **6**, the first support member **420** may be disposed adjacent to the lower end portion of the rear surface of the OCTA module **410** (or the cover panel **413** of the OCTA module **410**) (e.g., in the -y direction of FIG. **6**). According to an embodiment, an opening **421** configured to dispose the display driver IC **440** therein may be included in the first support member **420**. The first support member **420** according to an embodiment may be disposed to surround the display driver IC **440** on all sides to protect the display driver IC **440** from an external impact.

(97) According to an embodiment, a length **L1** of the first support member **420** in the first direction (e.g., the x-axis direction of FIG. **6**) may be equal to a length **L2** of the cover panel **413** in the first

direction. According to an embodiment, the first support member **420** may cover the entire area of the lower end portion of the rear surface of the cover panel **413** in the first direction. That is, the first support member **420** may be disposed extending from the center of the lower end portion of the cover panel **413** to opposite ends in the first direction.

(98) According to an embodiment, a length **L3** of the first support member **420** in the second direction (e.g., the y-axis direction of FIG. 6) may be longer than a height **L4** of the round shape of the corners of the cover panel **413**. According to an embodiment, the first support member **420** may cover in the second direction the cover panel **413** from the lower end portion (or an area adjacent to the lower end portion) of the rear surface of the cover panel **413** up to an area beyond at least the rounded shapes (or chamfered shapes) of the corners of the cover panel **413**. According to an embodiment, the first support member **420** may cover in the second direction the cover panel **413** from the lower end portion (or an area adjacent to the lower end portion) of the rear surface of the cover panel **413** at least up to an area beyond an area in which the display driver IC **440** is disposed.

(99) FIG. 7 is a view illustrating a state in which the second support members **430**, the panel flexible circuit film **450**, and the FPCB **460** are disposed on the rear surface of the display module **400** of FIG. 6 according to an embodiment of the disclosure.

(100) Referring to FIG. 7, the panel flexible circuit film **450** may be disposed adjacent to the lower end portion (e.g., the -y direction of FIG. 7) of the rear surface of the OCTA module **410** (or the cover panel **413** of the OCTA module **410**). According to an embodiment, the panel flexible circuit film **450** may be disposed to cover at least a partial area of the rear surface of the OCTA module **410** and at least a partial area of the rear surface of the first support member **420**. According to an embodiment, the panel flexible circuit film **450** may include the display driver IC **440** configured to control the OCTA module **410**, and the display driver IC **440** may be disposed between the rear surface of the OCTA module **410** and the flexible circuit films, that is, within the opening **421** of the first support member **420** disposed on the rear surface of the OCTA module **410**.

(101) According to an embodiment, at least the partial area of the first support member **420** may be exposed outside the panel flexible circuit film **450** when the display module **400** is viewed from the rear surface of the display module **400**. According to an embodiment, at least the partial area of the first support member **420** may have an area greater than that of the panel flexible circuit film **450** within a predetermined distance from the lower end portion of the display module **400**. For example, the length **L1** of the first support member **420** in the first direction (e.g., the x-axis direction of FIG. 7) is greater than a longest length **L5** of the panel flexible circuit film **450** in the first direction.

(102) According to an embodiment, a pair of second support members **431** and **432** may be disposed to cover opposite ends of the rear surface of the first support member **420**, respectively. According to an embodiment, the second support members **431** and **432** may each cover at least the partial area of the first support member **420** exposed outside the panel flexible circuit film **450** (i.e., the area that is not covered by the panel flexible circuit film **450**).

(103) According to an embodiment, the FPCB **460** may be disposed adjacent to the lower end portion of the rear surface of the OCTA module **410** (or the cover panel **413** of the OCTA module **410**). According to an embodiment, the FPCB **460** may be disposed to cover at least a partial area of the rear surface of the OCTA module **410** and at least a partial area of the rear surface of the panel flexible circuit film **450**. According to an embodiment, the FPCB **460** may be electrically connected to the OCTA module **410** (or the display panel **412** of the OCTA module **410**) via the panel flexible circuit film **450** and may control the OCTA module **410** via the flexible circuit film (or the display driver IC **440** included in the flexible circuit film).

(104) FIG. 8A is a cross-sectional view of the display module **400** of FIG. 7 taken in A-A' direction according to an embodiment of the disclosure, FIG. 8B is a cross-sectional view of the display module **400** of FIG. 7 taken in B-B' direction according to an embodiment of the disclosure, and

FIG. 8C is a cross-sectional view of the display module **400** of FIG. 7 taken in C-C' direction according to an embodiment of the disclosure.

(105) Referring to FIGS. 8A to 8C, the OCTA module **410** may include a transparent plate **411**, a display panel **412**, and a cover panel **413**, and the display panel **412** and the cover panel **413** may be sequentially stacked on the rear surface of the transparent plate **411** (e.g., the surface in the $-z$ direction in FIG. 8A).

(106) According to an embodiment, at least a portion of the display panel **412** may be visually recognized on the front surface (e.g., the surface in the $+z$ -direction surface in FIG. 8A) of the display module **400** (or the electronic device **300**) through the transparent plate **411** made of a transparent material. According to an embodiment, the transparent plate **411** may transmit at least one kind of light displayed on the display panel **412** and may protect the display panel **412** from an external physical impact. The transparent plate **411** may be made of at least one of glass and a polymer material such as polycarbonate (PC), polymethyl methacrylate (PMMA), polyimide (PI), polyethylene terephthalate (PET), or polypropylene terephthalate (PPT). According to an embodiment, the transparent plate **411** may include a multilayer structure of various materials.

(107) According to an embodiment, the display panel **412** covering the rear surface of the transparent plate **411** may include a plurality of layers. According to an embodiment, the display panel **412** may include a base substrate, a thin-film transistor (TFT) layer provided on the base substrate, and/or a pixel layer (or organic light-emitting layer) configured to receive a signal voltage from the thin-film transistor layer. The thin-film transistor layer may include an active layer, a gate insulating layer, a gate electrode, an interlayer insulating layer, a source electrode, and/or a drain electrode and may transmit signals necessary for driving the pixel layer. The pixel layer may include a plurality of display elements, for example, light-emitting diodes. The pixel layer may be defined as an area in which a plurality of organic display elements provided on the thin film transistor layer are disposed. According to an embodiment, the display panel **412** may further include a polarization member layer configured to transmit light corresponding to a predetermined direction, a thin-film encapsulation layer configured to encapsulate a pixel layer, and a back film configured to support a base substrate.

(108) According to an embodiment, the cover panel **413** covering the rear surface of the display panel **412** may include a plurality of layers. The cover panel **413** may be disposed under the display panel **412** to protect the display panel **412** and may include various components for facilitating bonding to other components of the display module **400**. According to an embodiment, the cover panel **413** may include a black layer (e.g., an embossed layer), a support layer (e.g., a sponge layer), a tape layer, a digitizer and/or a heat dissipation member layer (e.g., a copper (Cu) layer or a graphite layer). The black layer and/or the support layer may absorb an impact to the display panel **412**. In addition, the black layer may block light generated from the display panel **412** or light entering from the outside. The digitizer may detect a touch or hovering input made by using a stylus on the display module **400**. The heat dissipation member layer may dissipate heat generated from the display panel **412** and may include graphite and/or a copper film.

(109) According to an embodiment, in order to mutually bond the transparent plate **411** and the display panel **412**, the display panel **412** and the cover panel **413**, and respective layers included in these plates, an adhesive member (or a gluing agent or an adhesive) (not illustrated) may be used. The adhesive member may include, for example, a double-sided adhesive film, a pressure-sensitive adhesive (PSA), an optically clear adhesive (OCA) film, or an optically clear adhesive resin (OCR).

(110) Referring to FIG. 8A, the first support member **420**, the panel flexible circuit film **450**, and the second support members **430** may be disposed on the rear surface of the cover panel **413**. According to an embodiment, the first support member **420** may be disposed between the panel flexible circuit film **450** and the cover panel **413** or between the second support members **430** and the cover panel **413**. According to an embodiment, the first support member **420** may fill an empty

space which may occur due to a bending portion **451** of the panel flexible circuit film **450** or the display driver IC **440** (see FIG. **8C**). The function and shape (e.g., the thickness) of the first support member **420** will be described in detail with reference to FIG. **8C**.

(111) According to an embodiment, the panel flexible circuit film **450** may be disposed adjacent to the lower end portion (e.g., the $-y$ direction of FIG. **8A**) of the display module **400** (or the electronic device **300**) to cover the rear surface of the cover panel **413**. According to an embodiment, the panel flexible circuit film **450** may electrically connect the display panel **412** (or at least one layer included in the display panel **412**) and the FPCB **460** of the electronic device **300** to each other.

(112) According to an embodiment, the panel flexible circuit film **450** may include a bending portion **451** extending toward the front surface of the display panel **412** in an area adjacent to the lower end portion of the display module **400**, and the bending portion **451** may be electrically connected to the display panel **412** on the front surface of the display panel **412**. According to an embodiment, the bending portion **451** may extend toward the front surface of at least one layer included in the display panel **412** in an area adjacent to the lower end portion of the display module **400** and may be electrically connected to at least one layer included in the display panel **412** on the front surface of the layer.

(113) According to an embodiment, the second support members **430** may each cover at least a partial area of the rear surface of the first support member **420**. According to an embodiment, the second support members **430** may each cover at least a partial area of the first support member **420** exposed outside the panel flexible circuit film **450** or at least a portion of an area not covered by the flexible circuit film in the rear surface of the first support member **420**. According to an embodiment, a thickness $w1$ of the second support members **430** may be equal to a thickness $w2$ of the flexible circuit film. Accordingly, the second support members **430** may fill empty spaces between the first support member **420** and the front housing **310** which may occur due to the flexible circuit film.

(114) Referring to FIG. **8B**, the first support member **420** may cover the entire rear surface (e.g., the surface in the $-z$ direction in FIG. **8B**) of the cover panel **413** up to the opposite ends of the rear surface of the cover panel **413** in a first direction (e.g., the x -axis direction of FIG. **8B**). That is, the first support member **420** may be disposed extending from the center of the lower end portion of the cover panel **413** to the opposite ends in the first direction. According to an embodiment, the length of the first support member **420** in the first direction may be the same as the length of the cover panel **413** in the first direction, and the opposite ends of the first support member **420** in the first direction may be arranged to match the opposite ends of the cover panel **413** in the first direction.

(115) Referring to FIG. **8C**, the first support member **420** may fill an empty space between the panel flexible circuit film **450** and the cover panel **413** and/or an empty space between the second support members **430** and the cover panel **413** that may occur due to the bending portion **451** of the panel flexible circuit film **450** and/or the display driver IC **440**. According to an embodiment, a thickness $w3$ of the first support member **420** may be equal to or greater than a thickness $w4$ of the display driver IC **440** and may be smaller than the diameter D of the bending portion **451** of the panel flexible circuit film **450**. Accordingly, the first support member **420** may compensate for an empty space that may occur due to the bending portion **451** of the panel flexible circuit film **450** and/or the display driver IC **440**.

(116) FIG. **9** is a view illustrating impacts according to drop of an electronic device according to an embodiment of the disclosure.

(117) Damage caused by impact when the electronic device **300** is dropped may occur mainly in the thin-film encapsulation layer of the display panel **412** of the OCTA module **410**, and when there is an empty space around the thin-film encapsulation layer, the impact caused by the drop have a greater effect. Referring to FIG. **9**, when the electronic device **300** is dropped, the

horizontal/vertical progression of waves caused by a primary impact meets at the point of a secondary impact, so a greater impact may be delivered to the thin-film encapsulation layer of the OCTA module **410**.

(118) According to an embodiment, since the panel flexible circuit film **450** extending to the corners of the OCTA module **410** is used in the display module **400** in which the display driver IC **440** supporting an operating frequency of 120 Hz is embedded, an empty space around the thin-film encapsulation layer of the display panel **412** may come close to a corner of the OCTA module **410**, which may be in disadvantageous in terms of damage to the electronic device **300** due to a drop. Accordingly, in the case of the display module **400** of the electronic device **300** according to an embodiment of the disclosure, in order to suppress damage due to a drop, the area of the first support member **420** may be extended to corners of the OCTA module **410**, and the second support members **430** may be additionally disposed so that the empty spaces inside the electronic device **300** can be reduced.

(119) FIG. **10** is a table showing, in comparison, impact distributions (or stress contours) due to the secondary impacts of falling of FIG. **9** in a drop simulation of electronic devices according to an embodiment of the disclosure.

(120) Part (a) of FIG. **10** shows an electronic device **300-1** including a display module (e.g., the display module **400'** of FIG. **5A**) in which a 60 Hz display driver IC and a small-area panel flexible circuit film are embedded, part (b) of FIG. **10** shows an electronic device **300-2** including a display module in which a large-area panel flexible circuit film **450** is embedded as a 120 Hz display driver IC **440** is used, and part (c) of FIG. **10** shows an electronic device **300-3** according to an embodiment of the disclosure in which the area of the first support member **420** is increased and the second support members **430** are additionally disposed in the electronic device of part (b) of FIG. **10**. The electronic device **300-1** of part (a) of FIG. **10** may correspond to the electronic device **300a** of FIG. **4A**, and the electronic device **300-3** of part (c) of FIG. **10** may correspond to the electronic device **300c** of FIG. **4C**.

(121) Referring to FIG. **10**, in the case of the electronic device **300-2** of part (b) of FIG. **10** including the display module in which the 120 Hz display driver IC **440** and the large-area panel flexible circuit film **450** are embedded, the impact caused by a drop is increased by 12.5% compared to the electronic device **300-1** of part (a) of FIG. **10**. However, in the electronic device **300-3** of part (c) of FIG. **10** according to an embodiment of the disclosure (i.e., the electronic device **300-3** in which the area of the first support member **420** is increased and the second support members **430** are additionally disposed in the electronic device **300-2** of part (b) of FIG. **10**), it is possible to achieve an improvement in terms of the impact by a drop compared to the electronic device **300-2** of part (b) of FIG. **12** since the impact caused by a drop is increased 0.5% compared to the electronic device **300-1** of part (a) of FIG. **10**.

(122) FIGS. **11A**, **11B**, and **11C** are diagrams illustrating a process of assembling a display module of the electronic device according to various embodiments of the disclosure.

(123) FIG. **11A** is a view illustrating assembling the first support member **420** and the second support members **430** to the OCTA module **410** according to an embodiment, FIG. **11B** is a view illustrating assembling the panel flexible circuit film **450** and the FPCB **460** to the display module **400** of FIG. **11A** according to an embodiment, and FIG. **11C** is a view illustrating assembling the front housing **310** to the display module **400** of FIG. **11B** according to an embodiment.

(124) Referring to FIG. **11A**, the first support member **420** may be assembled adjacent to the lower end portion of the rear surface of the OCTA module **410**, that is, the rear surface of the cover panel **413** of the OCTA module **410**, and a pair of second support members **430** may be assembled adjacent to opposite ends of the rear surface of the first support member **420**. According to an embodiment, the area of the pair of second support members **430** may be smaller than that of the first support member **420**.

(125) Referring to FIG. **11B**, the panel flexible circuit film **450** may be assembled over the rear

surface of the first support member **420** and the rear surface of the OCTA module **410**, and the FPCB **460** may be assembled over the rear surface of the panel flexible circuit film **450** and the rear surface of the OCTA module **410**. According to an embodiment, the display driver IC **440** embedded in the panel flexible circuit film **450** may be disposed to be accommodated in the opening **421** of the first support member **420**. According to an embodiment, the panel flexible circuit film **450** may be disposed not to overlap the second support members **430** (i.e., not to cover the second support members **430**).

(126) Referring to FIG. **11C**, the front housing **310** may be assembled to the rear surface of the display module **400**. According to an embodiment, the display module **400** may be arranged to be seated on at least one step provided in the front housing **310**. According to an embodiment, the second support members **430** may each fill an empty space that occurs between the first support member **420** and the front housing **310** due to the panel flexible circuit film **450**.

(127) As described above, an electronic device (e.g., the electronic device **101** of FIG. **1**, the electronic device **200** of FIGS. **2A** and **2B**, or the electronic device **300** of FIG. **3**) according to an embodiment may include a housing defining at least a portion of an exterior of the electronic device, and a display module seated on the housing. The display module may include a transparent plate defining at least a portion of the front surface of the electronic device, a display panel at least partially visually recognized from the front surface of the electronic device through the transparent plate, a panel flexible circuit film including a bending portion disposed under the rear surface of the display panel and extending toward the front surface of the display panel in an area adjacent to a first side surface of the electronic device, wherein the bending portion is electrically connected to the display panel on the front surface of the display panel, and a first support member disposed between the rear surface of the display panel and the panel flexible circuit film, wherein at least a partial area of the first support member may have an area larger than the area of the panel flexible circuit film within a predetermined distance from the first side surface.

(128) According to an embodiment, at least the partial area of the first support member may be exposed outside the panel flexible circuit film when the display module is viewed from the rear surface of the display module, and the first support member may have a first length in a first direction that is greater than the longest length of the panel flexible circuit film in the first direction.

(129) According to an embodiment, the display module may further include at least one second support member disposed under the first support member, and the second support member may cover at least the partial area of the first support member exposed outside the panel flexible circuit film.

(130) According to an embodiment, the second support member may have a thickness that is equal to that of the panel flexible circuit film.

(131) According to an embodiment, the display module may further include a cover panel disposed between the rear surface of the display panel and the first support member, and the first support member may be configured to cover a rear surface of the cover panel from one end to another end of the cover panel in a first direction.

(132) According to an embodiment, the display module may further include a cover panel disposed between the rear surface of the display panel and the first support member, and the first support member may have a length in a first direction which is equal to a length of the cover panel in the first direction.

(133) According to an embodiment, the display module may further include a cover panel disposed between the rear surface of the display panel and the first support member, and the corners of the display panel and the cover panel may have a round shape, and the first support member may cover at least the round shape of the cover panel.

(134) According to an embodiment, the bending portion of the panel flexible circuit film may have a diameter greater than the thickness of the first support member.

(135) According to an embodiment, the display module may further include a flexible printed circuit board (FPCB), and the panel flexible circuit film may electrically connect the display panel and the flexible printed circuit board to each other.

(136) According to an embodiment, the display module may further include a display driver IC (DDI) disposed between the rear surface of the display panel and the panel flexible circuit film, the display driver IC may be electrically connected to the panel flexible circuit film to control driving of the display module, and the first support member may include an opening configured to accommodate the display driver IC therein.

(137) According to an embodiment, the display module may further include a display driver IC (DDI) disposed between the rear surface of the display panel and the panel flexible circuit film, and the first support member may have a thickness equal to or greater than a thickness of the display driver IC.

(138) According to an embodiment, the display module may further include a display driver IC (DDI) disposed between the rear surface of the display panel and the panel flexible circuit film, and the panel flexible circuit film and the display driver IC may constitute a chip-on-film (COF).

(139) According to an embodiment, the display driver IC may support an operating frequency of 120 Hz.

(140) According to an embodiment, the housing may include a first step and a second step, and the transparent plate may be seated on the first step, and the display panel, the panel flexible circuit film, and the first support member may be seated on the second step.

(141) According to an embodiment, the first side surface of the electronic device may be a lower end portion of the electronic device.

(142) As described above, a display module (e.g., the display module **160** of FIG. **1**, the display **201** in FIGS. **2A** and **2B**, or the display module **400** in FIG. **3**) of an electronic device (e.g., the electronic device **101** of FIG. **1**, the electronic device **200** of FIGS. **2A** and **2B**, or the electronic device **300** of FIG. **3**) according to an embodiment may include a display panel, a panel flexible circuit film including a bending portion disposed under the rear surface of the display panel and extending toward the front surface of the display panel in an area adjacent to a first side surface of the electronic device, wherein the bending portion is electrically connected to the display panel on the front surface of the display panel, and a first support member disposed between the rear surface of the display panel and the panel flexible circuit film. At least the partial area of the first support member may be exposed outside the panel flexible circuit film when the display module is viewed from the rear surface of the display module.

(143) According to an embodiment, the display module may further include at least one second support member disposed under the first support member, the second support member may cover at least the partial area of the first support member exposed outside the panel flexible circuit film, and the second support member may have a thickness that is equal to that of the panel flexible circuit film.

(144) According to an embodiment, the display module may further include a cover panel disposed between the rear surface of the display panel and the first support member, and the first support member may be configured to cover a rear surface of the cover panel from one end to another end of the cover panel in a first direction.

(145) According to an embodiment, the bending portion of the panel flexible circuit film may have a diameter greater than the thickness of the first support member.

(146) According to an embodiment, the display module may further include a display driver IC (DDI) disposed between the rear surface of the display panel and the panel flexible circuit film, the display driver IC may be electrically connected to the panel flexible circuit film to control driving of the display module, the first support member may include an opening configured to accommodate the display driver IC therein, and the first support member may have a thickness equal to or greater than a thickness of the display driver IC.

(147) In the above-described specific embodiments of the disclosure, components included in the disclosure have been expressed in singular or plural terms according to the presented specific embodiments. However, the singular or plural expressions are selected appropriately for the situation presented for convenience of description, and the disclosure is not limited to a singular constituent element or plural constituent elements. A constituent element expressed in a plural form may be configured with a singular element, or a constituent element expressed in a singular form may be configured with plural elements.

(148) While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a housing defining at least a portion of an exterior of the electronic device; and a display disposed on the housing, wherein the display comprises: a transparent plate defining at least a portion of a front surface of the electronic device, a display panel at least partially visually recognized from the front surface of the electronic device through the transparent plate, a panel flexible circuit film comprising a bending portion disposed under a rear surface of the display panel and extending toward a front surface of the display panel in an area adjacent to a first side surface of the electronic device, the bending portion being electrically connected to the display panel on the front surface of the display panel, a first support member disposed between the rear surface of the display panel and the panel flexible circuit film, at least a partial area of the first support member having an area larger than an area of the panel flexible circuit film within a predetermined distance from the first side surface, and a second support member disposed under the rear surface of the display panel, and wherein the second support member has a thickness corresponding to a thickness of the panel flexible circuit film.
2. The electronic device of claim 1, wherein at least the partial area of the first support member is exposed outside the panel flexible circuit film when the display is viewed from the rear surface of the display, and wherein the first support member has a length in a first direction that is greater than a longest length of the panel flexible circuit film in the first direction.
3. The electronic device of claim 2, wherein the second support member covers at least the partial area of the first support member exposed outside the panel flexible circuit film.
4. The electronic device of claim 1, wherein the display further comprises: a cover panel disposed between the rear surface of the display panel and the first support member, and wherein the first support member covers a rear surface of the cover panel from one end to another end of the cover panel in a first direction.
5. The electronic device of claim 1, wherein the display further comprises: a cover panel disposed between the rear surface of the display panel and the first support member, and wherein the first support member has a length in a first direction that is equal to a length of the cover panel in the first direction.
6. The electronic device of claim 1, wherein the display further comprises: a cover panel disposed between the rear surface of the display panel and the first support member, wherein corners of the display panel and the cover panel have a round shape, and wherein the first support member covers at least the round shape of the cover panel.
7. The electronic device of claim 1, wherein the bending portion of the panel flexible circuit film has a diameter greater than a thickness of the first support member.
8. The electronic device of claim 1, wherein the display further comprises: a flexible printed circuit board (FPCB), and wherein the panel flexible circuit film electrically connects the display panel to the flexible printed circuit board.

9. The electronic device of claim 1, wherein the display further comprises: a display driver integrated circuit (IC) disposed between the rear surface of the display panel and the panel flexible circuit film, wherein the display driver IC is electrically connected to the panel flexible circuit film and configured to control driving of the display, and wherein the first support member comprises an opening in which the display driver IC is disposed.
10. The electronic device of claim 1, wherein the display further comprises: a display driver integrated circuit (IC) disposed between the rear surface of the display panel and the panel flexible circuit film, and wherein the first support member has a thickness equal to or greater than a thickness of the display driver IC.
11. The electronic device of claim 1, wherein the display further comprises: a display driver integrated circuit (IC) disposed between the rear surface of the display panel and the panel flexible circuit film, wherein the panel flexible circuit film and the display driver IC constitute a chip-on-film (COF), and wherein the display driver IC supports an operating frequency of 120 hertz (Hz).
12. The electronic device of claim 1, wherein the housing comprises a first step and a second step, wherein the transparent plate is disposed on the first step, and wherein the display panel, the panel flexible circuit film, and the first support member are disposed on the second step.
13. The electronic device of claim 1, wherein the first side surface of the electronic device is a lower end portion of the electronic device.
14. A display of an electronic device, the display comprising: a display panel; a panel flexible circuit film including a bending portion disposed under a rear surface of the display panel and extending toward a front surface of the display panel in an area adjacent to a first side surface of the electronic device, the bending portion being electrically connected to the display panel on the front surface of the display panel; a first support member disposed between the rear surface of the display panel and the panel flexible circuit film, at least a partial area of the first support member being exposed outside the panel flexible circuit film when the display is viewed from the rear surface of the display; and a second support member at least partially covering the rear surface of the display panel, wherein the second support member has a thickness corresponding to a thickness of the panel flexible circuit film.
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